

Find Minimum in rotated array using Binary Search -

	Strt				Mid				End	While start < end
Index	0	1	2	3	4	5	6	7	8	
nums [] array	1	2	3	4	5	6	7	8	9	sorted but not rotated
rotate by 1 element	9	1	2	3	4	5	6	7	8	If mid < end Pivot point (Min) is in the left half including mid.
rotate by 2 elements	8	9	1	2	3	4	5	6	7	
rotate by 3 elements	7	8	9	1	2	3	4	5	6	
rotate by 4 elements	6	7	8	9	1	2	3	4	5	
rotate by 5 elements	5	6	7	8	9	1	2	3	4	If mid > end Pivot point (Min) is in the right half excluding mid
rotate by 6 elements	4	5	6	7	8	9	1	2	3	
rotate by 7 elements	3	4	5	6	7	8	9	1	2	
rotate by 8 elements	2	3	4	5	6	7	8	9	1	
Rotate by 9 elements	1	2	3	4	5	6	7	8	9	Full rotation

Facts about binary search on rotated array -

- In rotated array min and max elements are always next to each other.
- Pivot point is the index of the minimum element in the array. (pivot = min)

In below example -

- The minimum element here is 1
- The index of minimum element is 3
- This means that the array rotated by 3 elements or 3 times.

Index	0	1	2	3	4	5	6	7	8
nums [] array	7	8	9	1	2	3	4	5	6

- When a mid is calculated, the rotated array gets split in to two halves, left and right, having below characteristics -
 - only right-half is sorted. (ex. arrays rotated by 1, 2, 3 elements in above table)
 - both left and right are sorted and mid points to min or max element. (ex. arrays rotated by 4, 5 elements in above table)
 - only left-half is sorted. (ex. arrays rotated by 6, 7, 8 elements in above table).
 - Pivot Point is always in the un-sorted right or left half or at the mid.
- Binary search logic -
 - While $start < end$
 - If $mid > end$, pivot is in the right half.
 - If $mid < end$, pivot is in the left half, including mid.
 - When loop ends, both start and end points to min.